

Inverse Kaluza-Klein Scale Symmetry

Document I

An Alternative Solution Framework for the
Schrödinger Equation via
Scale-Projected Identity Decomposition

VSEPR-SIM Theoretical Division

Development Day 64

Abstract

The standard Schrödinger equation operates on a wavefunction $\Psi(\mathbf{r}, t)$ defined over classical 3D space plus time. This document proposes an alternative framework in which the wavefunction is reinterpreted as the *scale-projected shadow* of a fuller identity structure $\mathcal{I}$ defined over an extended scale ladder $\{S_0, S_1, \dots, S_D\}$. The standard solution is recovered at the S_3 – S_4 projection; corrections at higher scales produce measurable residuals that map onto dark-sector and quantum-gravity phenomenology. The framework does not modify the Schrödinger equation's algebra; it reinterprets what the wavefunction *is* and why it succeeds and fails where it does.

Contents

§1 The Standard Framework and Its Projection Assumption

1.1 The Schrödinger Equation as Usually Stated

The time-dependent Schrödinger equation is:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \hat{H} \Psi(\mathbf{r}, t) \quad (1)$$

where $\hat{H}$ is the Hamiltonian operator, $\mathbf{r} \in \mathbb{R}^3$, and Ψ is complex-valued. The Born rule assigns probability density:

$$\rho(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2 \quad (2)$$

This framework is extraordinarily predictive at the scale S_3 – S_4 (spatial/binding + time-ordered). It fails or requires patching at:

- Very high energy / short distance: ultraviolet divergences
- Many-body regimes: exponential Hilbert space scaling
- Gravitational coupling: no consistent quantum gravity
- Cosmological scale: dark matter and dark energy residuals
- Measurement: wavefunction collapse has no agreed dynamics

Remark 1.1. These failure modes are not random. Under the present framework they are systematic projection losses: physics that exists at scales beyond S_4 but is not captured by the $\Psi(\mathbf{r}, t)$ representation.

1.2 The Implicit Projection Assumption

The standard equation contains a hidden assumption:

$$\boxed{\Psi(\mathbf{r}, t) \equiv P_{S_3 \cup S_4}(\mathcal{I})} \quad (3)$$

That is, the wavefunction *is* the identity state. This is the assumption the present framework relaxes. Instead:

$$\Psi(\mathbf{r}, t) = P_{S_3 \cup S_4}(\mathcal{I}) \quad \text{with} \quad \mathcal{I} \supset P_{S_3 \cup S_4}(\mathcal{I}) \quad (4)$$

The residual is:

$$\mathcal{R}_{S_3} = \mathcal{U}_{S_3}(\mathcal{I}) - \mathcal{E}_{S_3}(\Psi) \quad (5)$$

where $\mathcal{U}_{S_3}$ is the full unprojected identity content at scale S_3 and $\mathcal{E}_{S_3}$ is the classical expectation recovery.

§2 Identity Structure and Scale Ladder

2.1 Scale Ladder Definition

Definition 2.1 (Scale entry). Each scale level S_n is a triple:

$$S_n = (D_n, M_n, L_n) \quad (6)$$

where D_n is the degree of freedom / dimensional permission added at scale n , M_n is the associated mediator or force carrier, and L_n is the observational limit (the scale below which S_n becomes relevant).

Scale	Meaning	Mediator	Observational limit
S_0	Binary existence $E \in \{0, 1\}$	—	Planck?
S_1	Linear vector permission	—	Sub-nuclear
S_2	Relational / angular	Gluons (color)	Nuclear
S_3	Spatial / binding structure	Photons (EM)	Atomic
S_4	Time-ordered transformation	W, Z bosons	Weak
S_5	Mass-energy curvature	Graviton (hypothetical)	Cosmological
S_D	Dark-scale permission w	X_D (unknown)	L_{EM}

Table 1: Scale ladder. Time t is the sequencing parameter of $\mathcal{I}(t)$, not a scale level itself.

2.2 Identity Bundle and Identity Anchor

A particle is not a point. It is an identity bundle:

$$\mathcal{I}_i(t) = \{c_{i1}, c_{i2}, \dots, c_{im}\} \quad (7)$$

where each component c_{ik} carries a scale-level tag.

The identity anchor (generalized center) is:

$$\mathcal{A}_i(t) = \frac{\sum_k \alpha_{ik} c_{ik}(t)}{\sum_k \alpha_{ik}} = \frac{\boldsymbol{\alpha}_i^\top \mathbf{C}_i}{\boldsymbol{\alpha}_i^\top \mathbf{1}} \quad (8)$$

The observed 3D position is only the projection:

$$\vec{r}_i^{\text{obs}} = P_{3D}(\mathcal{A}_i) \quad (9)$$

The fuzz radius measures internal spread at the observed scale:

$$r_f^2 = \frac{\sum_k \alpha_k \left\| P_{3D}(c_k) - \vec{r}^{\text{obs}} \right\|^2}{\sum_k \alpha_k} \quad (10)$$

Definition 2.2 (Fuzz ratio).

$$\Phi_S = \frac{r_f}{\ell_S} \quad (11)$$

where ℓ_S is the characteristic length of scale S .

$$\Phi_S \ll 1 \implies \text{point-like at scale } S$$

$$\Phi_S \sim 1 \implies \text{fuzz-zone}$$

$$\Phi_S > 1 \implies \text{non-point-defined at scale } S$$

§3 The Alternative Wavefunction

3.1 Extended State Space

In the standard framework, the state space is:

$$\Psi : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C} \quad (12)$$

In the extended framework, the full identity state lives in:

$$\Psi_{\text{full}} : \mathbb{R}^3 \times w \times \mathbb{R} \rightarrow \mathbb{C} \quad \Psi_{\text{full}} = \Psi(x, y, z, w; t) \quad (13)$$

where w is the dark-scale coordinate. The observed wavefunction is the projection:

$$\Psi_{\text{obs}}(\mathbf{r}, t) = \int \Psi_{\text{full}}(\mathbf{r}, w, t) K_{\text{EM}}(w) dw \quad (14)$$

with $K_{\text{EM}}(w)$ the electromagnetic projection kernel.

3.2 The Extended Schrödinger Equation

Replace $\hat{H}$ with the extended Hamiltonian:

$$\hat{H}_{\text{eff}} = \hat{H}_{S_3} + \hat{H}_{S_4} + \hat{H}_w \quad (15)$$

where $\hat{H}_w$ is the dark-scale Hamiltonian term. The extended equation becomes:

$$\boxed{i\hbar \frac{\partial}{\partial t} \Psi_{\text{full}}(\mathbf{r}, w, t) = \hat{H}_{\text{eff}} \Psi_{\text{full}}(\mathbf{r}, w, t)} \quad (16)$$

Integrating out w with kernel $K_{\text{EM}}(w)$ recovers (??) exactly when $\hat{H}_w$ is negligible, plus a correction term:

$$i\hbar \frac{\partial \Psi_{\text{obs}}}{\partial t} = \hat{H}_{S_3} \Psi_{\text{obs}} + \underbrace{\int \hat{H}_w \Psi_{\text{full}} K_{\text{EM}}(w) dw}_{\delta \hat{H}_w[\Psi_{\text{obs}}]} \quad (17)$$

Remark 3.1. $\delta \hat{H}_w$ is the residual Hamiltonian term invisible to EM projection. It encodes exactly the physics that the standard equation misses. In regimes where $\hat{H}_w \approx 0$, standard QM is recovered without modification.

3.3 Cancellation and Hidden Active Structure

A critical property of the extended framework is that observable channel cancellation does not imply internal inactivity.

Define the projected observable I_j and hidden intensity H_j for component j :

$$I_j = \boldsymbol{\alpha}^\top \mathbf{c}_j \quad (18)$$

$$H_j = \mathbf{c}_j^\top \mathbf{W} \mathbf{c}_j \quad (19)$$

The hidden-active condition:

$$\boxed{I_j = 0, \quad H_j > 0} \quad (20)$$

Example: neutron-like charge cancellation.

$$I_{\text{charge}} = +\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0 \quad (21)$$

$$H_{\text{charge}} = \left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 = \frac{2}{3} \neq 0 \quad (22)$$

The particle appears charge-neutral (zero observable channel) but has nonzero internal charge intensity. This structure is hidden from the K_{EM} projection but present in $\hat{H}_w$.

This generalizes to:

Observable channel	Hidden-active mechanism
Electric charge	Quark charge cancellation
Color charge	Color neutrality (confinement)
Spin	Singlet pairing
Linear momentum	Bound-state internal motion
Angular momentum	$L + S = J = 0$ nuclear ground states
Magnetic moment	Quenched orbital contributions
Defect pairs	Frenkel / Schottky charge balance
Mechanical stress	Residual stress in balanced structures

§4 Projection Failure and Exactness

4.1 Many-to-One Projection

Definition 4.1 (Projection non-injectivity). A projection P_S is non-injective when:

$$P_S(\mathcal{I}_a) = P_S(\mathcal{I}_b) \quad \text{while} \quad \mathcal{I}_a \neq \mathcal{I}_b \quad (23)$$

Theorem 4.1 (Exactness failure). *When P_S is non-injective:*

$$P_S^{-1}(\mathcal{O}_S) \text{ is not unique.} \quad (24)$$

Exact reconstruction from observation at scale S is impossible.

Sketch. Two distinct identity states $\mathcal{I}_a \neq \mathcal{I}_b$ produce identical observations $\mathcal{O}_S$. No measurement at scale S can distinguish them. A unique inverse map would require information not present at scale S . $\square$ $\square$

4.2 Destructive Inverse Packing

Packing is the forward process:

$$\mathbf{C}^{(n)} \xrightarrow{\Pi_n} \mathbf{I}^{(n+1)} \quad (25)$$

The inverse (annihilation / separation):

$$\mathbf{I}^{(n+1)} \xrightarrow{\mathcal{A}} \mathbf{C}_{\text{out}}^{(n)} \quad (26)$$

necessarily *destroys or transforms* the original packed identity.

Proposition 4.1. *Complete hidden-variable maps require destructive inverse packing. Therefore exact mapping cannot preserve the system being mapped.*

This connects to the quantum no-cloning theorem and to the irreversibility of measurement. It is not an epistemological limitation; it is a structural consequence of packing.

§5 Entanglement as a Relation-Particle

5.1 The Relation Object

Two entangled particles A and B share a relation object:

$$\mathcal{R}_{AB} = [0, 0, 0, 0 \mid \mathbf{T}_{AB}] \quad (27)$$

The first four entries are spacetime displacement channels:

$$\Delta x = \Delta y = \Delta z = \Delta t = 0 \quad (28)$$

The trinary relation vector:

$$\mathbf{T}_{AB} \in \{-1, 0, +1\}^m \quad (29)$$

carries the real correlational content.

Remark 5.1. $[0, 0, 0, 0]$ means *no spacetime displacement*, not *no physical existence*. The relation object is real, deterministic, and non-spatiotemporal.

5.2 Pair Decomposition and Measurement

The joint identity decomposes as:

$$\mathcal{I}_{AB} \rightarrow \mathcal{I}_A + \mathcal{I}_B + \mathcal{R}_{AB} \quad (30)$$

Measurement outcomes are scale-projected with the relation object:

$$O_A = P_a(\mathcal{I}_A, \mathcal{R}_{AB}) \quad (31)$$

$$O_B = P_b(\mathcal{I}_B, \mathcal{R}_{AB}) \quad (32)$$

The correlation function:

$$E(a, b) = \langle O_A O_B \rangle \quad (33)$$

reproduces quantum-mechanical predictions because $\mathcal{R}_{AB}$ is a fixed structure in identity-space, not a propagating signal.

§6 Dark Matter as Higher-Scale Projection Residual

6.1 The Projection Gap

$$\rho_{\text{dark}}(x, y, z; t) = P_G(\rho) - P_{\text{EM}}(\rho) \quad (34)$$

In kernel form, integrating over the dark coordinate w :

$$\rho_{\text{dark}}(x, y, z; t) = \int \rho(x, y, z, w; t) [K_G(w) - K_{\text{EM}}(w)] dw \quad (35)$$

$K_G(w)$ is the gravitational projection kernel (couples to all scales); $K_{\text{EM}}(w)$ is the electromagnetic projection kernel (truncates at L_{EM}).

6.2 Dark-Scale Entry

The dark-scale level in the ladder:

$$S_D = (w, X_D, L_{\text{EM}}) \quad (36)$$

where w is the dark-scale degree of freedom, X_D is the (presently unknown) dark mediator, and L_{EM} marks where electromagnetic projection fails.

Central claim:

$$\boxed{\text{dark matter} = \text{gravitational signature of an unresolved higher-scale permission}} \quad (37)$$

This is distinct from standard extra-dimensional models:

Standard: dimension first $\rightarrow$ dark behavior

Inverse KK: force-scale residuals first $\rightarrow$ inferred dimension

§7 Effective Proper Time with Dark-Scale Correction

7.1 Classical and Quantum Proper Time

Classical relativistic proper time:

$$d\tau_{4D} = dt \sqrt{1 - \frac{v^2}{c^2} + \frac{2\Phi_G}{c^2}} \quad (38)$$

Recent optical ion clock work establishes:

$$\hat{\tau} \equiv \tau(\hat{x}, \hat{p}) \quad (39)$$

Proper time becomes quantum-linked; $\hat{\tau}$ is an observable with spread $R_\tau = \hat{\tau} - \langle \hat{\tau} \rangle$.

7.2 Dark-Scale Extension

Extend to:

$$d\tau_{\text{eff}} = dt \sqrt{1 - \frac{v^2}{c^2} + \frac{2\Phi_G}{c^2} + \chi_w(w)} \quad (40)$$

where $\chi_w(w)$ is the dark-scale metric correction term.

In differential form:

$$d\tau_{\text{eff}} = d\tau_{4D} + d\tau_w \quad (41)$$

The full proper-time identity structure:

$$\mathcal{I}_\tau = (\hat{x}, \hat{p}, \hat{\tau}, H_c, H_m, w) \quad (42)$$

where H_c is the clock Hamiltonian and H_m is the motional Hamiltonian. The clock-motion relation object:

$$\mathcal{R}_{cm} = [0, 0, 0, 0 \mid \mathbf{T}_{cm}] \quad (43)$$

Remark 7.1. $\chi_w(w)$ remains formally undefined here. Constraining it requires either: (a) a specific model for $\hat{H}_w$, or (b) comparison with clock-discrepancy data in strong gravitational fields or extreme velocities where 4D relativity is insufficient.

§8 Recovery of Standard Quantum Mechanics

The critical consistency check: the standard Schrödinger equation must be exactly recovered in the appropriate limit.

Proposition 8.1 (Standard QM recovery). *When $\hat{H}_w \approx 0$ (dark-scale Hamiltonian negligible) and $\chi_w(w) \approx 0$ (proper-time correction negligible), the extended framework reduces exactly to:*

$$\Psi_{\text{obs}}(\mathbf{r}, t) \approx \Psi(\mathbf{r}, t) \quad (44)$$

$$i\hbar \partial_t \Psi = \hat{H}_{S_3} \Psi \quad (45)$$

which is (??).

The framework is therefore a *conservative extension*. No existing prediction is contradicted; corrections appear only where the extended terms are non-negligible: strong gravity, cosmological scales, and dark-sector phenomenology.

§9 Open Problems

1. **Form of $K_{\text{EM}}(w)$ and $K_G(w)$.** These projection kernels determine the magnitude of all dark-sector corrections. They are presently unconstrained.
2. **Form of $\hat{H}_w$.** Without a specific dark-scale Hamiltonian the theory has no new quantitative predictions.
3. **Form of $\chi_w(w)$.** The dark-scale metric correction in (??) needs a physical derivation or at minimum a phenomenological ansatz.
4. **Trinary relation vector $\mathbf{T}_{AB}$.** The space $\{-1, 0, +1\}^m$ must be given a dynamics. How does $\mathbf{T}_{AB}$ evolve under local operations?
5. **Packing operator Π_n .** The map from $\mathbf{C}^{(n)}$ to $\mathbf{I}^{(n+1)}$ is stated as an axiom. Its explicit form at each scale level is needed.
6. **Connection to existing field theory.** The extended equation (??) must be Lorentz-covariant. The dark coordinate w must respect causality or explain why it does not.

§A Notation Summary

Symbol	Meaning
$\mathcal{I}$	Full identity state
P_S	Projection onto scale S
$\mathcal{R}_S$	Residual at scale S : $\mathcal{U}_S(\mathcal{I}) - \mathcal{E}_S(\mathcal{O}_S)$
$S_n = (D_n, M_n, L_n)$	Scale entry: DOF, mediator, observational limit
w	Dark-scale degree of freedom
$\mathcal{A}_i$	Identity anchor (eq. ??)
r_f	Fuzz radius (eq. ??)
Φ_S	Fuzz ratio
Π_n	Packing operator
$\mathcal{R}_{AB}$	Relation-particle object (eq. ??)
$\mathbf{T}_{AB}$	Trinary relation vector
$K_G(w), K_{\text{EM}}(w)$	Gravitational / EM projection kernels
$\chi_w(w)$	Dark-scale metric correction
$d\tau_{\text{eff}}$	Effective proper time (eq. ??)